Portfolio

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DSCI 609: Data Analytics for Social Sciences

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In this capstone project, I demonstrate my ability to use 6 different data analysis techniques to gain further insight on real-world data using R. For each technique, I used data from the General Social Survey (GSS) which is a nationally representative interview survey for the United States adult population that collects social, political, and a behavioral data among many other topics. Each analysis included its own hypothesis which was then either confirmed or negated. The results will be shown afterwards, accompanied by the complete R script and data frames used.

Independent Samples T-test

I wanted to figure out if being male or female had a significant correlation with supporting restrictions in obtaining guns. This dataset included the results of individuals that either favored or opposed a potential law which would require a person to obtain a police permit before he or she could buy a gun from the years 2012 – 2022. My hypothesis was that females would be more supportive of this law in comparison to males. The Pew Research Center (2017) found that women are more likely than men to support gun control measures. This article touched upon the use of background checks and licensing requirements. The RAND Corporation (2023) reported consistent differences in public opinion on gun policies, with women expressing greater support for restrictive firearm regulations. The NPR/Ipsos (2022) poll further reinforces this trend, showing that most women, even including female gun owners, supported stricter gun laws such as permit requirements and age limits for purchasing firearms. Given the information, it was safe to hypothesize that more females would be supportive of this law in comparison to men. The

dependent variable was the support of this law, and the independent variable was the two genders. Since the participants only belonged to one gender group and were not paired in any way, a dependent samples t-test would not have worked. The independent samples t-test allowed me to test my hypothesis that women would show higher support for gun permitting laws compared to men.

Using an independent samples t-test to see if there were significant differences between male and females when it came to the support of a law which would require a person to obtain a police permit before he or she could buy a gun, results indicated that females (mean = 1.24, standard deviation = 0.429) were more supportive of this law from the years 2012-2022 than males (mean = 1.36, standard deviation = 0.479), $t(11477) = -13.748$, $p < .001$, $d = -0.249$.

In the dataset where I made the values of the gun law column, favor and oppose, numeric; favor equaled 1 and oppose equaled 2. So, in this case, the lower mean and standard deviation indicated greater support. The results confirmed my hypothesis that women would support the restrictive gun law more than men. The difference was statistically significant which gives us the idea that the results weren't due to chance. The effect size was small; however, this was consistent with the research.

Links:

R Script

Dataset

Two-Way Anova

For the two-way ANOVA, I wanted to test the effect of religious affiliation and marital status on life satisfaction. My hypothesis was that those that were married and religious, would return answers that reflected more life satisfaction. I also believed that there would be an interaction effect, meaning the influence of religion on happiness would change depending on marital status. My hypothesis was based on research suggesting that both marital status and religious affiliation are associated with increased happiness, and that their effects may interact. Pew Research Center (2019) found that individuals who are actively religious report higher levels of happiness and greater well-being than those who are unaffiliated or inactive in their faith. Diener, Tay, and Myers (2011) discussed that there have been past studies that have found that religious folk have higher subjective well-being. An interview with Brad Wilcox in TIME (2024) further emphasized that marriage and religious commitment together are strong predictors of happiness. These findings informed my hypothesis that married, religious individuals would report the highest happiness levels. I believed that the effects of religion and marriage on happiness would not just be additional, but the data might show that it is much more interactive. I selected a two-way ANOVA because I wanted to test the main effects of two independent categorical variables, marital status and religion, on a continuous outcome variable, happiness. Additionally, I wanted to assess whether there was an interaction effect between these two predictors. A two-way ANOVA is the appropriate analysis when testing for differences in means across combinations of two categorical variables and whether their effects on the dependent variable are independent or interdependent.

A Tukey's HSD post hoc analysis revealed that the happiness of married folk was significantly greater than individuals that were never married, widowed, separated, and divorced.

Two-Way Anova results indicated a significant main effect for marriage, $F(4, 16927) = 297.330$, $p < .001$, $\eta^2 = 0.065$ which means that there was a significant difference across marital groups. and also a significant main effect for religion, $F(12, 16927) = 2.606$, $p = 0.002$, $\eta^2 = 0.002$. The effect size is small with only little variance. What I should have done was group all the religions together and compare that to non-religious folk to really know the difference. What I did here was only see the happiness levels between different religious groups. The interaction between the two was not statistically significant, $F(40, 16927) = 1.167$, $p = .217$, $\eta^2 = 0.003$. Being both married and religious does not produce a significantly different effect on happiness beyond the individual contributions of each. My hypothesis was only partially supported.

Links:

R Script

Dataset

Multiple Regression

I hypothesized that age would be negatively associated with self-rated health, such that older individuals would report lower health ratings. I also predicted that higher educational attainment would be associated with better self-rated health, and that race would also be a significant predictor, with racial disparities present in health outcomes. Together, I expected these variables to significantly predict variation in how people rate their own health. The dataset I used was from the years 1972 – 2022. My hypothesis was informed by research articles showing how age, education, and race are significant predictors of self-reported health. The Kaiser Family Foundation (2023) highlights persistent racial disparities in health outcomes

across the U.S., noting that Black, Hispanic, and Native American populations are more likely to report poor health compared to White individuals, due in part to systemic inequities in access to care and social determinants of health. The Nature article by Wojtyniak et al. (2024) examined self-reported health in patients with 13 different diseases and found that age negatively correlates with health-related quality of life, even after controlling for disease group. This supports the idea that older individuals, regardless of condition, tend to perceive their health more poorly. Lastly, Zajacova and Lawrence (2018) reviewed the strong link between educational attainment and health, noting that education affects health both directly and through broader social contexts like employment, behavior, and healthcare access. Together, these findings provide a solid rationale for expecting that age, education, and race all predict self-reported health.

Using a multiple regression analysis, Health was entered as the outcome variable, while Age, Race, and Education were entered as the simultaneous predictors. The overall model was significant, $F(25, 54050) = 295.2$, $R^2 = .120$, $p < 0.001$. Indicating that 12 percent of the variance in Health was accounted for by the three predictors (Adjusted $R^2 = 0.119$). The effect size is small, but it is reliable in predictability. Table 1 provided the individual predictor regression coefficients.

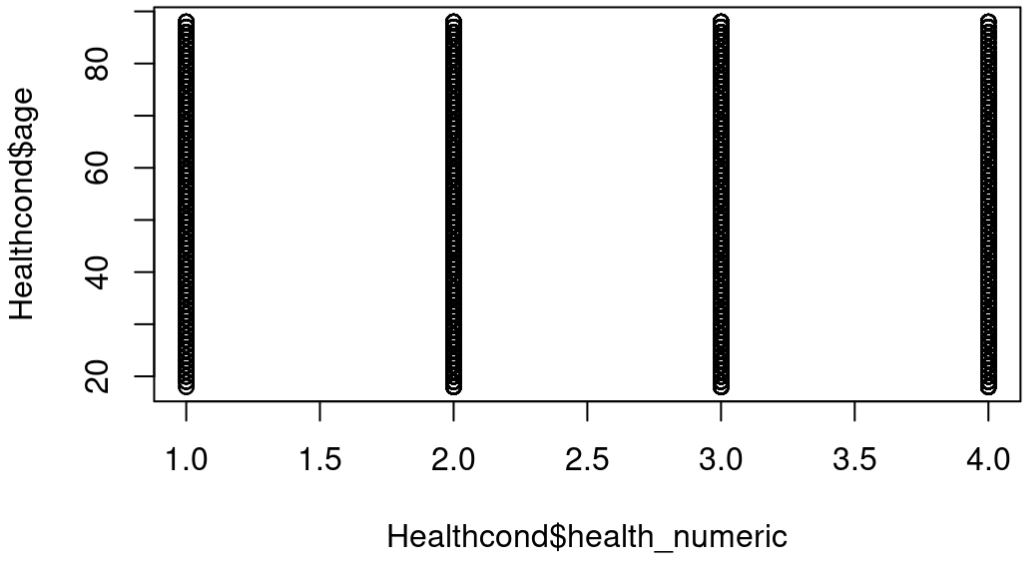
Table 1

Regression Coefficients for Multiple Regression

	Estimate	*SE*	*t*	*p*
age	-0.0095145	0.0002025	-46.982	< 2e-16
educ.n: No answer	0.1965876	0.1333947	1.474	0.140560
educ1 year of college	0.3416867	0.0996571	3.429	0.000607
educ10th grade	0.0155630	0.1002793	0.155	0.876667
educ11th grade	0.1009459	0.1000020	1.009	0.312769
educ12th grade	0.2919769	0.0991465	2.945	0.003232

educ1st grade	-0.0684091	0.1655533	-0.413	0.679451
educ2 years of college	0.3781572	0.0994590	3.802	0.000144
educ2nd grade	0.0056499	0.1221661	0.046	0.963113
educ3 years of college	0.4116464	0.1002346	4.107	4.02e-05
educ3rd grade	-0.1883266	0.1131871	-1.664	0.096148
educ4 years of college	0.5630891	0.0993724	5.666	1.47e-08
educ4th grade	-0.2488575	0.1113606	-2.235	0.025441
educ5 years of college	0.5365000	0.1006725	5.329	9.91e-08
educ5th grade	-0.2210682	0.1083767	-2.040	0.041374
educ6 years of college	0.5994511	0.1003494	5.974	2.33e-09
educ6th grade	-0.1175598	0.1035560	-1.135	0.256285
educ7 years of college	0.6330402	0.1026034	6.170	6.89e-10
educ7th grade	-0.0849531	0.1032855	-0.823	0.410792
educ8 or more years of college	0.6258194	0.1011840	6.185	6.26e-10
educ8th grade	-0.0375446	0.1004220	-0.374	0.708504
educ9th grade	-0.0205551	0.1008031	-0.204	0.838422
educNo formal schooling	-0.3545666	0.1207097	-2.937	0.003312
raceOther	0.0147302	0.0164398	0.896	0.370254
raceWhite	0.1556036	0.0098035	15.872	< 2e-16

Graph 1



Age was a strong and statistically significant negative predictor of self-rated health. This supports the hypothesis that older individuals tend to report lower health ratings. Educational attainment showed a general positive trend with self-rated health. It was noticeable that after 4 years of college, the coefficients were significantly positive. Individuals with more education were more likely to rate their health higher. This also aligns with the original hypothesis. Some lower education categories weren't all that significant, but for the most part, the pattern displayed an overall positive trend. Individuals identifying as White were significantly more likely to report higher health ratings than the reference group. The coefficient for the "Other" group wasn't statistically significant. This partially supports the hypothesis that racial disparities exist in self-reported health.

Links:

R Script

Dataset

Mediation

I hypothesized that job satisfaction would mediate the relationship between income and depression. I expected that individuals with higher income would report greater job satisfaction, which in turn would be associated with lower levels of depression. In other words, income would be indirectly associated with reduced depression through its influence on job satisfaction. My hypothesis was informed by research showing that income influences depression partially through job satisfaction. The American Psychological Association (2021) reported that workers facing financial insecurity often experience lower job satisfaction and elevated mental health concerns. Bowling, Eschleman, and Wang (2010) demonstrated that job satisfaction is closely tied to subjective well-being, reinforcing its potential mediating role between socioeconomic status and psychological outcomes. Patel et al. (2018) conducted a systematic review showing that income inequality is significantly associated with depression with factors such as work-

related stress and perceived lack of control being involved. These studies collectively support the use of mediation analysis to examine whether job satisfaction helps explain the link between income and depression.

Table 2

Mediation Analysis of Total, Direct, and Indirect Effects

			95% CI		
Effect	Path	Estimate	Lower	Upper	p
Total Effect	Income > Depression	0.0268	0.01	0.04	0.02
Direct	Income > Depression	0.0254	0.01	0.04	0.02
Indirect	Income > Job Satisfaction > Depression	0.0014	-0.0007	0.003	0.16

A Sobel test shows that the mediation analysis was not significant, $Z = 1.313$, $p = 0.189$.

The results showed that the total effect of income on depression was significant, $b = 0.0268$, $p < .001$, indicating that higher income was associated with lower depression scores. However, the results do not support my hypothesis that job satisfaction mediates the relationship between income and depression. While both income and job satisfaction are individually associated with depression, the statistical evidence did not support a mediation effect in this model.

Links:

R Script

Dataset

Multi-Level Modeling

My hypothesis was that higher levels of education would be associated with higher income, but that this relationship might differ across U.S. regions in the year 2024. I expected a positive relationship between education and income, and I hypothesized that region would moderate this relationship, so the strength of the association might vary depending on where a person lives. This hypothesis is supported by research from several data-driven sources. The U.S. Bureau of Labor Statistics (2023) highlights that individuals with higher levels of education consistently earn more on average than those with less education, clearly illustrating the direct financial benefit of educational attainment. The National Center for Education Statistics (2023) reports that young adults with bachelor's degrees or higher have substantially greater annual incomes than those with only a high school diploma. Research by Chetty et al. (2014) also shows that where a person lives plays a crucial role in economic mobility. Their analysis found substantial regional differences in income mobility and outcomes, even among individuals with similar levels of education. This suggests that geographic context can significantly moderate the relationship between education and income, supporting the use of a multi-level modeling approach that accounts for variation across regions.

First, a baseline model was created with income as the DV and no predictors, as well as a fixed and random intercept with region as the cluster variable. Results indicated that the clustering variable accounted for 1 percent of the variability in the data set (ICC = 0.01). The

variability in the group intercepts (SD = 0.24 and Residual SD = 2.64) was significantly different from the fixed intercept (M = 10.83 , SE = 0.10 , $p < 0.001$)

Next, a Fixed Effects model was fitted with Education as a fixed slope predictor on being able to predict Income. Results indicated that Education was a significant predictor. The Random Effects variability (SD = 0.179) and Residual variability (SD = 2.607) were both insignificant.

Next, a Random Effects model was fitted with Education as a random slope predictor on being able to predict Income. Results again indicated that Education (SD = 0.05) was still a significant predictor. The Random Effects for Intercept variability (SD = 0.96) and Residual variability (SD = 2.60) barely was reduced from the Fixed Effects model.

Education remained a significant predictor of income across both models. The Random Slopes model did NOT significantly improve fit over the Fixed Slopes model. So while Education was still a significant predictor, allowing its slope to vary by region didn't meaningfully improve prediction. This test does not support the hypothesis that region would have a significant impact on income, although it does show that education does. Part of this may be due to the dataset only involving one year rather than multiple years.

Links:

R Script

Dataset

Exploratory Factor Analysis

My hypothesis was that individuals would exhibit similar levels of confidence in authority figures across different institutions. I expected that trust in institutions would be relatively uniform and cluster together, reflecting a general disposition toward institutional trust. However, I also believed that there would be some outliers like the military or religion, suggesting different trust profiles across sectors like government, media, business, and religious institutions. In this dataset, I used the responses from the confidence in institutions section of the 2018 GSS survey. Pew Research Center (2019) report revealed that Americans perceive declining trust in institutions as a barrier to solving key societal issues. This decline tends to differentiate between types of institutions likw government vs. military vs. media. This suggests that trust is not uniformly applied but instead clusters into distinct domains. Zmerli and Newton (2011) provide a conceptual framework distinguishing three types of trust; social, political, and institutional. Their research suggests that institutional trust forms a unique dimension separate from interpersonal or generalized trust, with internal variation even among different institutions. Spadaro et al. (2020) emphasize that institutional trust can shape interpersonal trust. It highlights that trust in systems like the legal or governmental structure may influence trust in people. This research supports the idea that trust across domains may still form distinct, measurable constructs. This led to the usage of EFA to discover these factors.

The sampling adequacy was acceptable overall (KMO = 0.78), and our correlations among the items were demonstrated to be large enough for the factor analysis by a Bartlett's test ($X2(28) = 1823.64$, $p < 0.001$).

Eigenvalue reporting:

Two factors were extracted from the data accounting for 28% of the variance (Factor 1 = 0.16 and Factor 3 = 0.12). The Eigenvalue for Factor 1 was 1.26 and for Factor 3 it was 0.95.

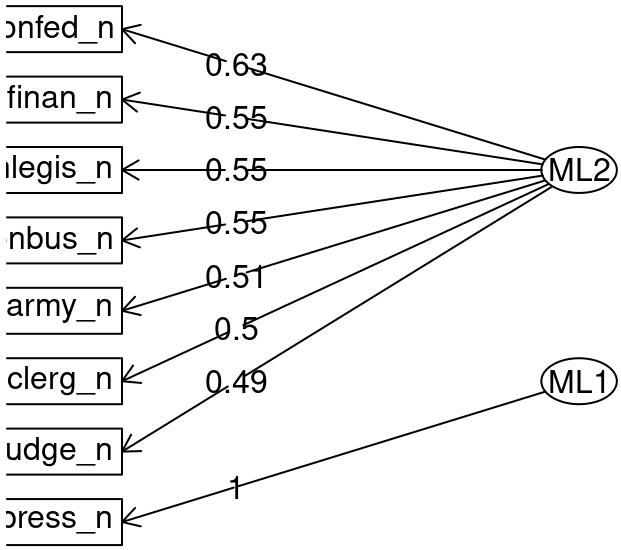
Table 3

Factor Loadings

	Factor 1	Factor 2
confinan_n	0.55	0.03
conbus_n	0.55	0.04
conclerg_n	0.50	-0.06
confed_n	0.63	-0.16
conpress_n	0.00	1.00
conjudge_n	0.49	0.14
conlegis_n	0.55	0.12
conarmy_n	0.51	-0.01

Graph 2

Confidence in Authority



Links:

R Script

Dataset

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